Cog2BOLD Effect Atlas

User guide to the `effect_atlas.py` app and the HTML report

Cog2BOLD documentation

October 2026

Abstract

The effect atlas asks a *trained* Cog2BOLD model a neuroscientist's question: *if this person's cognitive scores were different, where in the brain would the predicted activation change, in which direction, and by how much?* For every interesting combination of cognitive features it computes an *effect map*, finds the blobs of activation and deactivation, and reports each one as in a classic neuroimaging paper: peak MNI coordinates, anatomical label, functional network and cluster size. This guide explains how to drive the interactive app, how to read every part of the HTML report, and how to interpret the results correctly. All examples are *real* outputs from the default run `ds000030/pamret/ds000030_pamret_linear_identity_0_ridge_corr_gt_ctrl_meanoff`.

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1 What the effect atlas answers

A Cog2BOLD run is a function from a person's cognitive scores to a whole-brain predicted map (here the first-level `correct_gt_control` contrast of the paired-associate retrieval task, *pamret*). The effect atlas queries that function at carefully chosen profiles and subtracts the predictions.

The effect map Δ

For two cognitive profiles A and B,

$$\Delta(\text{voxel}) \;=\; \text{pred}(A)(\text{voxel}) \;-\; \text{pred}(B)(\text{voxel}).$$

Δ is in the units of the model's target maps. For this run those are z-statistic units of the subject-level `correct_gt_control` contrast. A positive Δ means profile A is predicted to have a *larger* contrast at that voxel than profile B.

The scan builds four kinds of combinations:

Kind	What is compared
single	One feature at its *high* vs *low* level (mean +1 SD vs mean −1 SD by default; 1 vs 0 for binary features such as diagnosis dummies), all other features held at their training mean. This is the *marginal effect* of the feature.
pairwise	The interaction of two features, HH − HL − LH + LL: how much the effect of one feature depends on the other. **Identically zero for linear predictors**, so it is skipped for them automatically and the report says so.
archetype	k-means clusters of the *real* training subjects (k chosen automatically by silhouette, 2–6). Each cluster centroid is contrasted with the mean profile: "what does the brain of this kind of person look like, relative to average?"
custom	Any two profiles you build with sliders in the app.

Each effect map is then thresholded, its connected blobs are found, and each blob is labelled.

The one rule: these maps describe the model, not the brain

An effect map shows what the *trained predictor* would change if a feature changed. It is exactly as trustworthy as the model's out-of-sample signal. A model that predicts held-out subjects no better than chance will still produce beautiful, "significant-looking" blobs: they are the model's fit to noise. **Always read the validity banner first** (Sections 3.2 and 4.1), and compare the model's LTO accuracy with its *empirical* permutation null, never with the nominal 0.25.

2 Quick start

Interactive app. From the repository root:

```
cd ~/Documents/Technion/Research/Cog2BOLD && source .venv/bin/activate
streamlit run apps/effect_atlas.py
```

The app opens on the default run (ds000030 → pamret → the meanoff run) in *effect* mode, so a first scan appears within seconds.

Report from the command line. For any trained run:

```
cog2bold effects --run-dir <run_dir> --mode bootstrap --n-boot 200
```

The report goes to `<run_dir>/effect_atlas_<mode>_<timestamp>/report.html`. For MLP runs use `-n-boot 50`: each bootstrap refit retrains the network, which takes about 2 s (linear refits take milliseconds). All command-line options are listed in Appendix A.

Which mode? Use *effect* mode to explore: it is instant and descriptive. Use *bootstrap* mode for anything you intend to report: it attaches an uncertainty estimate to every voxel and every ROI.

3 The app, tab by tab

3.1 Layout and the sidebar

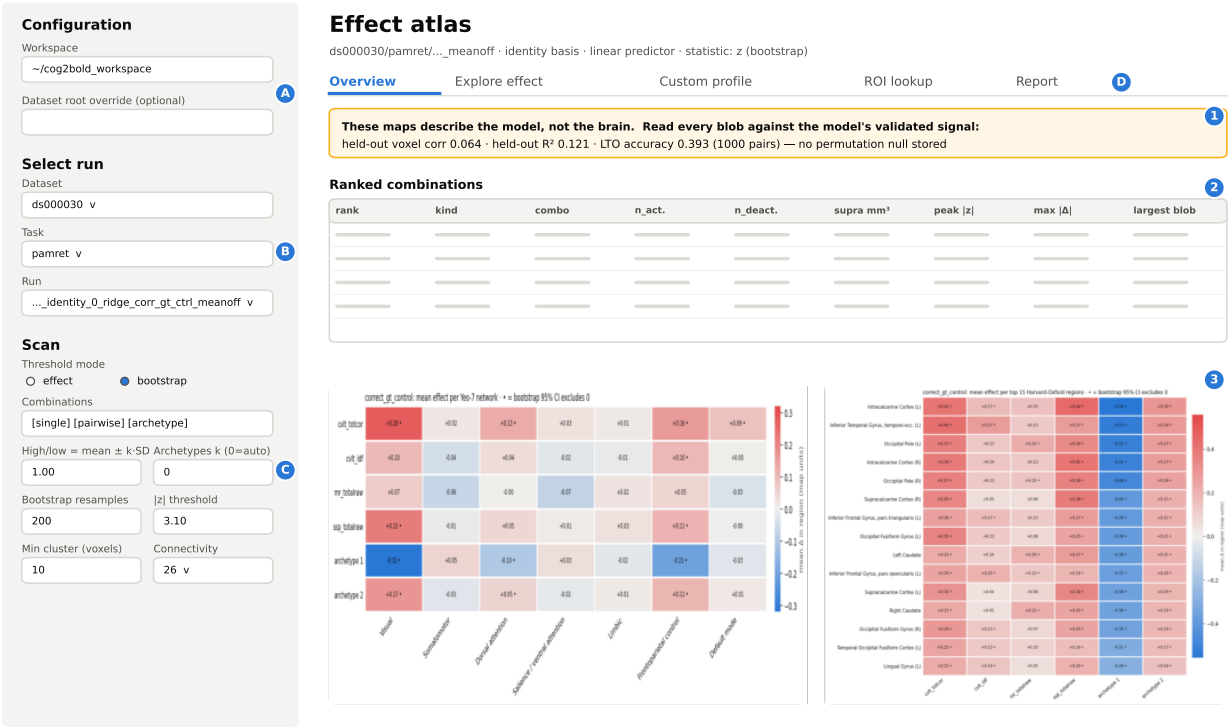


Figure 1: App layout on the default run (bootstrap mode, Overview tab). Letters mark the parts shared by every tab; numbers are explained in Section 3.2. The diagram is a faithful schematic of the app; the heatmaps are the real figures it shows for this run.

The sidebar (Figure 1) has three blocks:

- **A Configuration.** *Workspace*: where runs live (`$COG2BOLD_WORKSPACE`, default `~/cog2bold_workspace`). *Dataset root override*: leave blank; fill it only if a dataset moved since the run was trained.
- **B Select run.** A cascading picker, Dataset → Task → Run. Only fully trained runs (with `predictor.pkl` and `predictions.pkl`) are listed.
- **C Scan settings.** These decide what is computed (see the table below).
- **D Tabs.** Overview, Explore effect, Custom profile, ROI lookup, Report.

Control	Default	What it does / when to change it
Threshold mode	effect	*effect*: statistic = Δ. *bootstrap*: statistic = $z = \Delta/\mathrm{SD}_{\mathrm{boot}}$. Bootstrap scans start only when you press *Run bootstrap scan*.
Combinations	all three	Untick kinds you do not need. With many features, pairwise grows as $F(F-1)/2$.
High/low = mean $\pm k$·SD	1.0	Size of the feature change. Δ scales linearly with k for linear models.
Archetypes k	0 (auto)	Fix k if you want a specific number of subject types.
Pooled $\lvert\Delta\rvert$ percentile	99	*effect mode*: voxels with $\lvert\Delta\rvert$ above this percentile of $\lvert\Delta\rvert$ *pooled over all maps of the scan* survive. Pooling means a stronger combination keeps more voxels than a weaker one, so the ranking stays meaningful.
Bootstrap resamples	200 (linear) / 50 (MLP)	*bootstrap mode*: number of subject resamples. Use $\geq$200 for anything you report; small numbers inflate z.
$\lvert z\rvert$ threshold / FDR q	3.1 / off	*bootstrap mode*: voxelwise threshold, or Benjamini–Hochberg FDR.
Min cluster (voxels)	10	Clusters smaller than this are dropped. One ds000030 voxel is $3 \times 3 \times 4 = 36\,\mathrm{mm}^3$.
Connectivity	26	Which neighbouring voxels count as one cluster (26 as in FSL, 18 as in SPM, 6 is face-only).

3.2 Overview tab

1. **Validity banner.** The run's held-out voxel correlation, held-out R^2, and the accuracy of its most recent leave-two-out (LTO) evaluation. For the default run it reads: voxel correlation 0.064, R^2 0.121, LTO accuracy 0.393 over 1000 pairs, *no permutation null stored*. Without that null you cannot tell whether 0.393 is above chance for this evaluation design. Run `cog2bold lto -run-dir <run> -run-null` to get one.
2. **Ranked combinations.** One row per combination, ordered by total suprathreshold volume, then by $\|\Delta\|_2$ (Table 1 explains the columns).
3. **Heatmaps.** The "which feature moves which ROI" summary (Figure 2). Left: mean Δ in each Yeo-7 network. Right: the 15 Harvard–Oxford regions with the largest effects.

Below the heatmaps: the archetype profile table (Table 2) and an expander with a paper-ready methods paragraph.

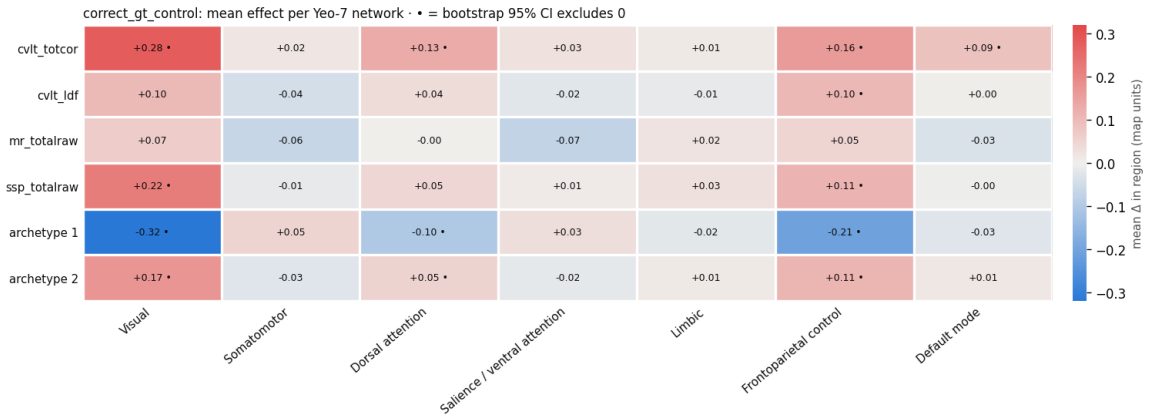


Figure 2: Mean effect per Yeo-7 network (default run, bootstrap mode). Rows are combinations, columns are networks; red = the combination increases the contrast there, blue = decreases. A dot (•) means the bootstrap 95% confidence interval of that network mean excludes 0. Example: `cvlt_totcor` +1 SD vs −1 SD raises the contrast in Visual (+0.28, CI [0.15, 0.40]), Frontoparietal control (+0.16, [0.07, 0.24]), Dorsal attention (+0.13, [0.03, 0.24]) and Default mode (+0.09, [0.01, 0.15]); its effects in Somatomotor, Salience/ventral attention and Limbic are not distinguishable from 0.

3.3 Explore effect tab

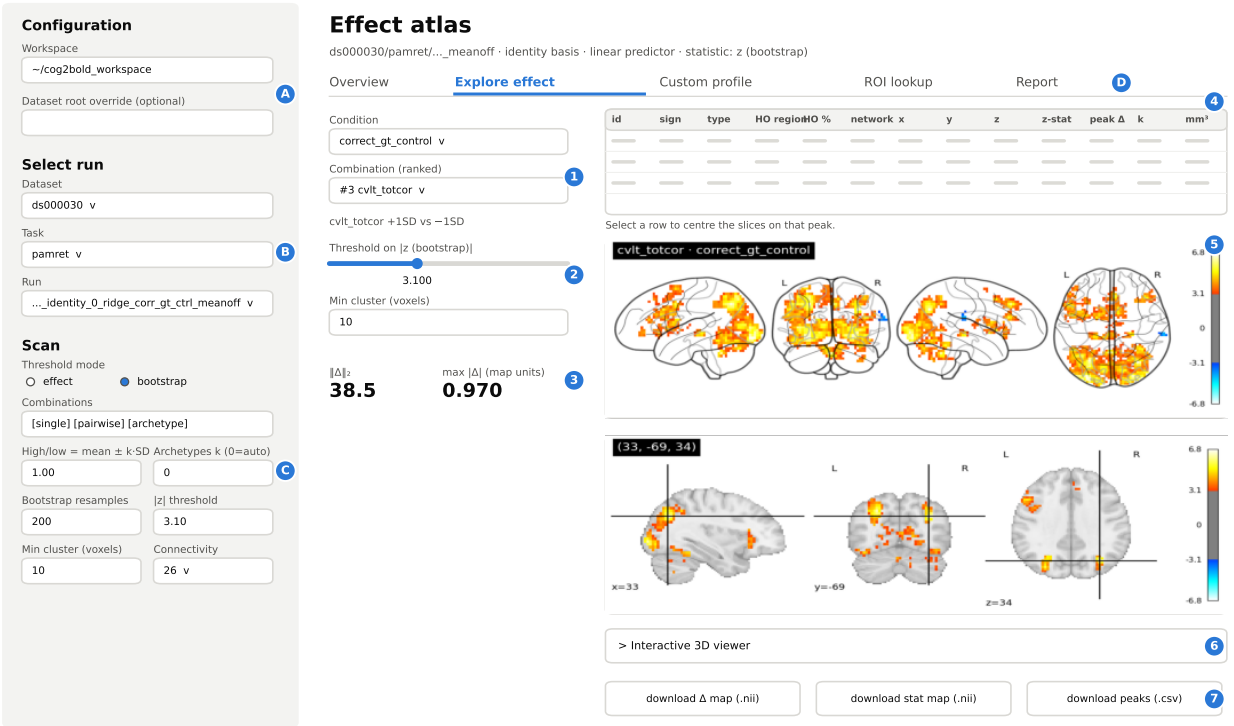


Figure 3: Explore effect tab, showing `cvlt_totcor` on the default run. The brain figures are the real ones for this effect.

1. **Pick the effect.** Condition (contrast) and combination; the list is in rank order. The line below repeats the definition ("cvlt_totcor +1SD vs −1SD").
2. **Live threshold.** Drag to re-threshold this map only; clusters, peaks and labels are recomputed instantly (no re-scan). The minimum cluster size can be changed here too.
3. **Effect size.** $\|\Delta\|_2$ (whole-map magnitude) and max $|\Delta|$, the largest voxel change in map units. Always read these next to z (Section 5.2).

4. **Peak table.** One row per peak (Table 4 lists the columns). *Click a row* to centre the slice views on that peak.

5. **Brain views.** A glass brain (all surviving clusters, projected) and orthogonal slices through the strongest peak, or through the row you selected.

6. **Interactive 3D viewer.** Click anywhere to move the crosshair; hover for values.

7. **Downloads.** The Δ map, the statistic map (both NIfTI, ready for FSLeyes or MRIcroGL) and the peak table (CSV).

3.4 Custom profile tab

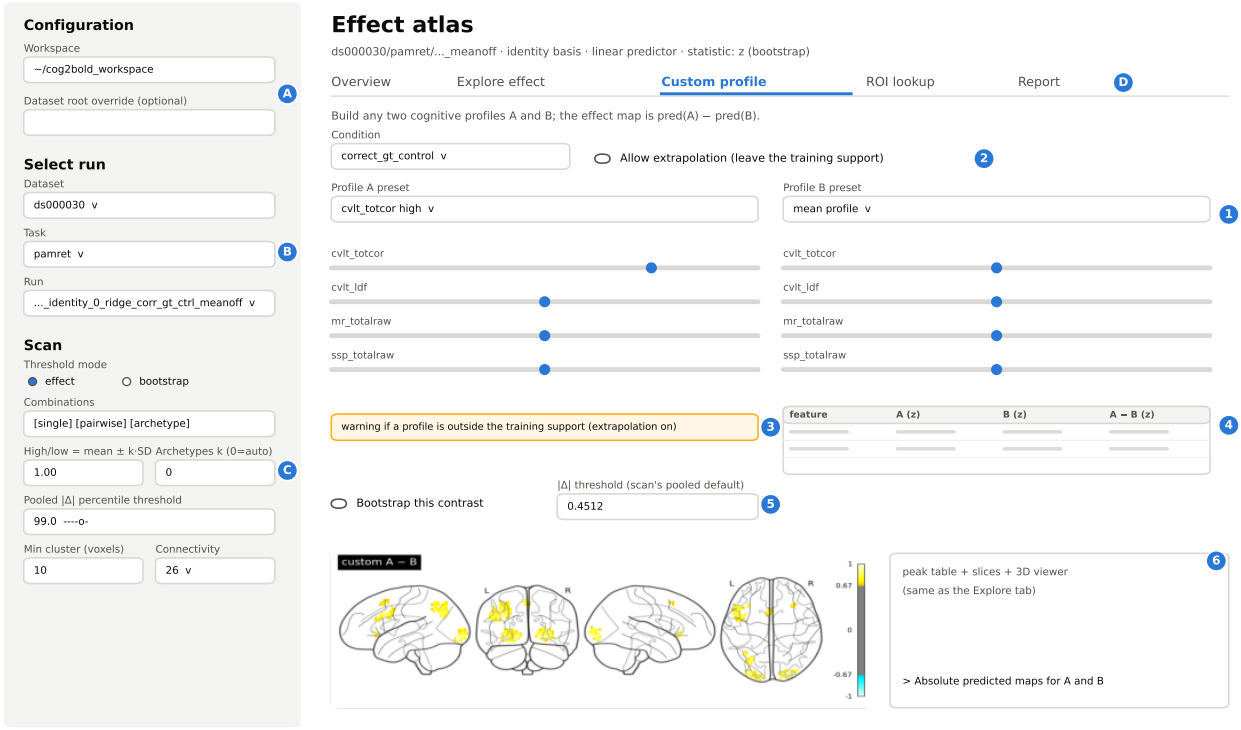


Figure 4: Custom profile tab. The glass brain is the real effect of the presets "cvlt_totcor high" (A) vs "mean profile" (B) on the default run.

1. **Profiles A and B.** Start from a preset, then fine-tune with the sliders. Presets: *mean profile*, f *high* and f *low* for every feature, and every archetype. In the f *high/low* presets the other features are set to their most plausible values given f, not to the population mean: when features correlate, "high on one, average on the rest" can be an unrealistic person.

2. **Allow extrapolation.** *Off (default)*: the sliders are limited to the joint plausible region of the training subjects (the 95% region of a Gaussian-copula model). Moving one feature narrows the range of the features correlated with it. *On*: any value in the training range is allowed.

3. **Out-of-support warning.** With extrapolation on, a warning appears when a profile is less typical than 95% of training subjects. The model never saw such a person, so treat the map as extrapolation.

4. **Profile table.** A and B as feature z-scores, and their difference.

5. **Threshold.** Effect mode uses the scan's pooled $|\Delta|$ threshold by default, so custom maps are comparable with the scanned ones. Tick *Bootstrap this contrast* for a z map of this specific A−B.

6 **Result.** Same views as the Explore tab, plus an expander with the *absolute* predicted maps of A and B (not differences).

3.5 ROI lookup and Report tabs



Figure 5: Left: ROI lookup for the Frontoparietal control network (real values for the default run, bootstrap mode). Right: the Report tab.

ROI lookup answers the reverse question: *I care about this region; which features move it?*

1 Choose an atlas (Yeo-7 networks or Harvard–Oxford regions), a region and a condition.

2 One bar per combination: the mean Δ over the model's voxels in that region. Red bars increase the contrast, blue bars decrease it. Whiskers are bootstrap 95% CIs (bootstrap mode only). A whisker crossing 0 means no reliable effect. In Figure 5, `cvlt_totcor` raises the frontoparietal-control contrast (CI clear of 0) while `mr_totalraw`'s CI crosses 0.

Report writes everything for the current scan:

3 How many combinations get figures in the HTML, whether to include NIfTI maps, and where to write: into the run directory, or to a temporary folder for download only.

4 *Build report*, then download the whole folder as a zip or just `report.html`.

4 The HTML report, section by section

The report is a single self-contained file: images are embedded, so you can e-mail it or open it offline. It is written next to tables and maps (Appendix B). The sections appear in this order.

4.1 Model card, validity banner, notes and methods

- **Model card**: dataset adapter, condition(s), basis, predictor and loss, the input features, the number of training subjects (158 here), and the threshold mode.
- **Validity banner** (yellow): the same numbers as the app's banner. When no permutation null is stored it says so explicitly: the nominal 25% chance level is an upper bound on the true chance accuracy, so it must not be used as the reference.
- **Notes**: anything the scan decided for you, e.g. "Pairwise interactions skipped: the predictor is linear", or the effect-mode threshold value.

- **Methods**: an auto-generated paragraph you can adapt for a paper. It states how Δ was built, the threshold, the cluster rules and the atlases, including the template caveat (Section 5.5).

4.2 Ranked combinations

rank	kind	combo	act.	deact.	supra mm^3	peak $\lvert z\rvert$	max $\lvert\Delta\rvert$	largest blob
1	archetype	archetype 1	13	10	251,496	11.1	1.18	Lateral Occipital Cortex, inferior division (L) (−)
2	archetype	archetype 2	10	13	251,496	11.1	0.63	Lateral Occipital Cortex, inferior division (L) (+)
3	single	cvlt_totcor	17	1	102,420	6.8	0.97	Lateral Occipital Cortex, superior division (R) (+)
4	single	ssp_totalraw	19	3	35,820	5.8	0.85	Intracalcarine Cortex (R) (+)
5	single	mr_totalraw	8	3	11,520	6.0	0.96	Occipital Pole (L) (+)
6	single	cvlt_ldf	9	1	11,448	5.1	0.73	Inferior Frontal Gyrus, pars opercularis (L) (+)

Table 1: Ranked combinations for the default run (bootstrap, 200 resamples, $\lvert z\rvert > 3.1$, $\geq$10 voxels). **act./deact.**: number of activation/deactivation clusters. **supra mm^3**: total volume of all surviving clusters (the ranking key). **peak** $\lvert z\rvert$: strongest voxel. **max** $\lvert\Delta\rvert$: largest voxel change in map units, *the effect size*. **largest blob**: label of the biggest cluster, (+) activation, (−) deactivation.

4.3 Heatmaps and archetype profiles

The same two heatmaps as the Overview tab (Figure 2), one pair per condition, followed by the archetype table:

archetype	n	cvlt_totcor	cvlt_ldf	mr_totalraw	ssp_totalraw
archetype 1	55	-1.02	-0.98	-0.99	-0.85
archetype 2	103	+0.54	+0.53	+0.53	+0.46

Table 2: Archetypes of the default run (k = 2 chosen by silhouette, 0.43). Values are the centroid's feature z-scores. Here the two archetypes are simply "below average on everything" (55 subjects) and "somewhat above average on everything" (103 subjects).

4.4 Per-combination sections

For the top combinations (30 by default), in rank order:

1. a heading with the definition and the condition, and a line with the threshold, $\|\Delta\|_2$ and $\max|\Delta|$;
2. a **glass brain**: the surviving clusters seen through a transparent brain from the left, front (y), right and top (z). Hot colours = activation (positive statistic), cold = deactivation;
3. **orthogonal slices** through the strongest peak, on the MNI template;
4. the **peak table**.

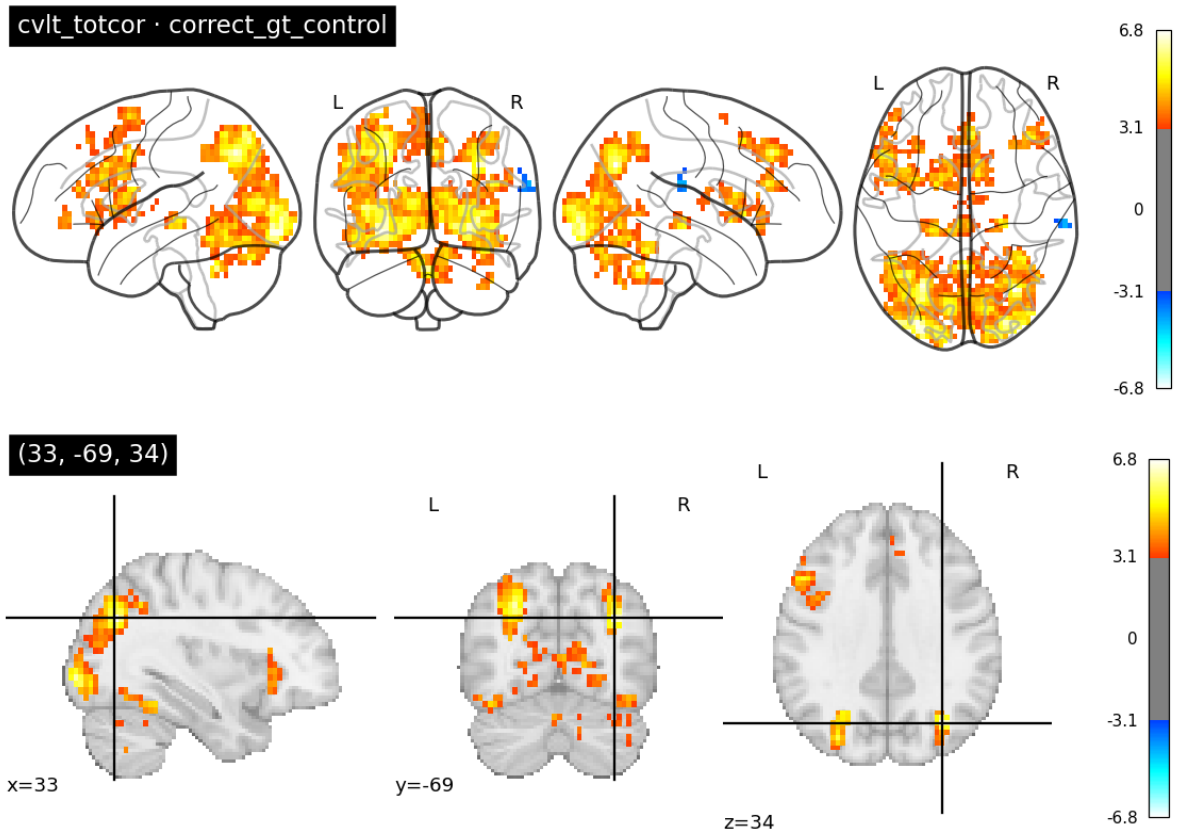


Figure 6: Report figures for `cvlt_totcor` on the default run. Top: glass brain. Bottom: slices through the strongest peak, MNI $(33, -69, 34)$, right superior lateral occipital cortex. The colour bar is in z units; the grey band is the sub-threshold range $|z| < 3.1$.

id	type	Harvard–Oxford region	HO %	net.	x	y	z	z-stat	peak Δ	k	mm^3
1	peak	Lateral Occipital Cortex, superior division (R)	52	Vis	33	−69	34	+6.81	+0.82	1851	66,636
	sub	↪ Lateral Occipital Cortex, inferior division (L)	27	Vis	−30	−90	−10	+6.67	+0.87		
	sub	↪ Lateral Occipital Cortex, superior division (L)	52	DorsAttn	−27	−66	42	+6.59	+0.89		
2	peak	Precentral Gyrus (L)	51	Cont	−48	3	42	+5.49	+0.71	206	7,416
	sub	↪ Middle Frontal Gyrus (L)	48	Cont	−51	18	34	+5.31	+0.62		
	sub	↪ Inferior Frontal Gyrus, pars triangularis (L)	24	Cont	−54	21	26	+4.89	+0.70		
3	peak	Paracingulate Gyrus (R)	31	Cont	3	33	42	+5.92	+0.61	161	5,796
	sub	↪ Superior Frontal Gyrus (R)	12	SalVentAttn	9	15	50	+4.98	+0.42		
	sub	↪ SMA (juxtapositional lobule) (L)	14	SalVentAttn	−9	−3	62	+4.05	+0.33		
4	peak	Left Caudate	25	–	−18	0	18	+4.97	+0.46	89	3,204
	sub	↪ Left Caudate	89	–	−9	6	10	+4.68	+0.47		
	sub	↪ Left Caudate	98	–	−12	15	6	+4.47	+0.39		

Table 3: The first 12 rows of the `cvlt_totcor` peak table (all activations). Sub-peaks (↪) are other local maxima of the same cluster, at least 8 mm from the main peak; size columns appear only on the main peak. The full table, with extra columns, is in `peaks.csv`.

Table 4: Peak-table columns (`peaks.csv` has all of them; the HTML and LaTeX tables show the main ones).

Column	Meaning
`cluster_id`	Cluster number; positive = activation, negative = deactivation. Ordered by size.
`sign`	*activation* (statistic > 0) or *deactivation* (statistic < 0).
`peak_type`	*peak* = cluster maximum; *sub* = secondary local maximum ($\geq$8 mm away).
`ho_region, ho_prob`	Harvard–Oxford region at the peak and its probability (%). Cortical names carry the hemisphere from the sign of x: (L) for $x < 0$, (R) for $x \geq 0$.

Column	Meaning
`ho_flag`	Empty if the label is direct. Otherwise: *white matter*, *nearest d mm* (the peak sits in an unlabelled voxel, so the nearest region within 4 mm is reported), *low probability* ($< 10\%$), or *no label*.
`schaefer_parcel, network`	Schaefer-200 parcel at the peak and its Yeo-7 network: Vis, SomMot, DorsAttn, SalVentAttn, Limbic, Cont (frontoparietal control), Default.
`x, y, z`	Peak location in MNI mm.
`peak_stat`	The statistic at the peak: z (bootstrap mode) or Δ (effect mode).
`peak_delta`	Δ at the peak, *always* in map units.
`n_voxels, volume_mm3`	Cluster size.
`cog_x/y/z`	Centre of gravity of the cluster (|statistic|-weighted).
`extent_*_mm`	Bounding-box size of the cluster along each axis.
`mean_delta`	Mean Δ over the cluster.
`abs_delta_share`	Share of the map's total $\lvert\Delta\rvert$ that falls inside this cluster.
`sign_consistency`	Bootstrap only: fraction of resamples in which Δ has the same sign as the full-data Δ (mean over the cluster). Close to 1 = robust direction.
`regions_covered`	The three Harvard–Oxford regions holding the most voxels of the cluster, with their share.

5 Interpreting the results

5.1 What an activation blob means

Because the model predicts a *contrast* map (`correct_gt_control`: correct retrieval > control), a blob is a statement about *individual differences in that contrast*:

How to read a blob

"*Activation* at (x, y, z) for feature f" means: according to the model, people higher on f show a *larger correct > control difference* there. It does *not* mean the task activates that region, nor that f "activates" it. "*Deactivation*" means a *smaller* difference for people higher on f.

The size of the change is `peak_delta`: for `cvlt_totcor` at the right lateral occipital peak it is +0.82, so going from -1 SD to $+1$ SD on the CVLT total raises the predicted subject-level contrast there by about 0.8 z-units.

5.2 Effect size versus z; effect versus bootstrap mode

A high z means stable, not big

The bootstrap $z = \Delta/\mathrm{SD}_{\mathrm{boot}}$ measures how *reproducible* an effect is across resamples of the training subjects. A negligible but very consistent Δ can reach a large z. On an MLP run of the same task, for example, max $\lvert\Delta\rvert$ was only 0.02 yet z reached 7. **Report Δ (`peak_delta`, `max |Δ|`) next to every z.**

- **Effect mode** is descriptive. The threshold is relative (a pooled percentile), so there are always some blobs: use it to see *where* the model puts its effects and how combinations compare. No significance is implied.
- **Bootstrap mode** adds uncertainty. The bootstrap refits only the predictor; the spatial basis is held fixed, so this is uncertainty *conditional on the basis*. The $\lvert z\rvert > 3.1$ threshold is voxelwise and uncorrected; use the FDR option for a corrected threshold.

- Ranking is by suprathreshold volume, which in bootstrap mode rewards *stability*. In Table 1 the archetypes rank first (251,496 mm^3) although their max $|\Delta|$ (1.18 and 0.63) is comparable to the single features: a contrast that moves all four features together is estimated very stably.

5.3 Worked example: `cvlt_totcor` on the default run

Reading Table 3 and Figures 2–6 together:

1. **Validity first.** The banner shows held-out $R^2 = 0.12$ and an LTO accuracy without a stored null. Every statement below is about the model and is provisional until the model's accuracy is shown to exceed its empirical null.
2. **Direction.** 17 activation clusters and 1 deactivation cluster: higher CVLT total is predicted to go with a larger retrieval contrast, almost everywhere the effect is reliable.
3. **Where.** Cluster 1 is very large (1,851 voxels, 66,636 mm^3). It spans bilateral lateral occipital cortex and occipital pole, which you can see from its sub-peaks and from `regions_covered`. Distinct frontal clusters: left precentral/middle frontal/IFG (cluster 2, frontoparietal-control network), paracingulate/pre-SMA (cluster 3), and left caudate (cluster 4). The network heatmap agrees: Visual, Frontoparietal control, Dorsal attention and Default have CIs excluding 0.
4. **How much.** Peak Δ of about 0.6–0.9 z-units per 2 SD of CVLT total.
5. **How robust.** `sign_consistency` ≈ 0.999 for these clusters: the direction holds in essentially every resample.

A suitable sentence: *"In the trained model, a 2-SD increase in CVLT total recall was associated with a larger correct>control retrieval contrast in bilateral lateral occipital cortex (peak MNI 33, −69, 34; $\Delta = +0.82$, bootstrap $z = 6.8$), left middle frontal/precentral gyrus (−48, 3, 42) and paracingulate cortex (3, 33, 42)."*

5.4 Archetypes

An archetype map contrasts a *type of real subject* with the average subject. For a linear model it equals the corresponding weighted sum of the single-feature maps, so it contains no new information beyond them. Its value is interpretive: it is a realistic profile. In this run the two archetypes lie on the same "everything low / everything high" axis, so their maps are the same pattern with opposite signs and identical z values. For nonlinear models (MLP) archetypes can differ genuinely from the sum of their parts.

5.5 Anatomical labels: what to trust

- **Probability.** `ho_prob` is the probability that the peak voxel belongs to the named region (Harvard–Oxford is probabilistic). Below 25% the label is uncertain; below 10% it is replaced by a white-matter or nearest-region label and flagged.
- **Large clusters span many regions.** Do not describe a 66,000 mm^3 cluster by its peak label alone. Use the sub-peaks and `regions_covered`.
- **Cerebellum and brainstem** are not covered by Harvard–Oxford cortical or Schaefer, so cerebellar peaks show *unlabelled* or *no label* (e.g. $(-9, -78, -26)$ for `cvlt_totcor`). Name them from their coordinates.
- **Template mismatch.** Both atlases are defined in FSL MNI152 (NLin6Asym), whereas the ds000030 and AOMIC maps are in MNI152NLin2009cAsym. The difference is a few millimetres: important at region borders, negligible for large clusters.

- **Network label vs anatomical label.** They come from different atlases and can disagree near borders (e.g. superior frontal gyrus in the Salience/ventral attention network). That is expected.

5.6 Checklist before reporting a finding

1. The model's LTO accuracy beats its *empirical* permutation null (not 0.25).
2. The scan is in bootstrap mode with $\geq$200 resamples (FDR for a corrected claim).
3. You quote Δ (map units) alongside z.
4. `sign_consistency` is close to 1 for the clusters you describe.
5. Large clusters are described through sub-peaks and `regions_covered`, not only their peak label.
6. In-scanner performance measures used as features are flagged as circular.
7. Effects of interest are not simply effects of `age` or other confounds: check them against a model that includes those features.

6 FAQ and troubleshooting

My run does not appear in the picker.
Only runs with both `predictor.pkl` and `predictions.pkl` are listed. Re-run `cog2bold train`.

"Could not load ..."
Usually the dataset (or its brain mask) is no longer at the path stored at training time. Set *Dataset root override* in the sidebar.

Where are the pairwise interactions?
Linear predictors have none (exactly zero); see the note in the Overview tab and the report.

No suprathreshold clusters.
Lower the threshold (Explore tab slider), lower the minimum cluster size, or check max $|\Delta|$: the model may predict almost no dependence on that feature.

Everything is significant.
Look at max $|\Delta|$ and at the validity banner. Large z with tiny Δ, or a model at chance, are not findings.

The bootstrap scan is slow.
It refits the predictor B times. Linear runs take milliseconds per refit; MLP runs take seconds. Lower *Bootstrap resamples* for exploration.

Why does the archetype rank first?
Multi-feature contrasts are estimated very stably; see Section 5.2.

A Command-line options

```
cog2bold effects --run-dir <run_dir> [options]
```

Option	Default	Meaning
`--mode`	effect	`effect` or `bootstrap`
`--combos`	single,pairwise,archetype	Combination kinds to scan
`--features`	all	Restrict to these features (comma-separated)
`--conditions`	all	Restrict to these conditions/contrasts
`--k-sd`	1.0	High/low = mean $\pm k$·SD
`--archetypes-k`	auto	Number of archetypes (default: silhouette, 2–6)
`--percentile`	99	Effect mode: pooled $\lvert\Delta\rvert$ percentile threshold
`--abs-threshold`	–	Effect mode: absolute $\lvert\Delta\rvert$ threshold (overrides percentile)
`--n-boot`	200	Bootstrap resamples
`--z-thr`	3.1	Bootstrap: voxel $\lvert z\rvert$ threshold
`--fdr-q`	–	Bootstrap: BH-FDR q (overrides `--z-thr`)
`--min-cluster-voxels`	10	Minimum cluster size
`--connectivity`	26	6, 18 or 26
`--min-peak-distance-mm`	8	Minimum distance between peaks of one cluster
`--max-subpeaks`	2	Sub-peaks listed per cluster
`--max-detailed`	30	Combinations with figures in the HTML
`--no-maps`	off	Do not write NIfTI maps
`--out-dir`	run dir	Output folder
`--seed`	run seed	Seed for archetypes and bootstrap

To produce reports for every trained run:

```
for d in $(find ~/cog2bold_workspace/runs -name predictor.pkl \
              -exec dirname {} \;); do
  [ -f "$d/predictions.pkl" ] || continue
  cog2bold effects --run-dir "$d" --mode bootstrap --n-boot 100
done
```

B Files in a report folder

File	Content
`report.html`	The self-contained report described in Section 4.
`summary.csv`	One ranked row per (condition, combination) (Table 1).
`peaks.csv`	Every peak with all columns of Table 4.
`peaks.md`	Peak tables in Markdown, one section per combination.
`peaks.tex`	Paper-ready peak table (`booktabs` + `longtable`).
`roi_effects_yeo7.csv`	Mean Δ per Yeo-7 network per combination (+ bootstrap CI).
`roi_effects_ho.csv`	Same for Harvard–Oxford regions.
`figures/`	All figures as PNG.
`maps/*_delta.nii.gz`	Δ map per effect (map units).
`maps/*_stat.nii.gz`	Statistic map (Δ or z).
`maps/*_clusters.nii.gz`	Cluster labels (+id activation, −id deactivation).
`scan_params.json`	Every setting of the scan, for reproducibility.